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 Studierichting: Informatica
 Oefeningenreeks 1D nr. 19.8

Opgave: $A(z) = z^4 - 5z^3 + 9z^2 - 15z + 18$

$$A(2) = 16 - 40 + 36 - 30 + 18 = 0$$

horner:

$$\begin{array}{r|rrrr|r} & 1 & -5 & 9 & -15 & 18 \\ 2 & \downarrow & 2 & -6 & 6 & -18 \\ \hline & 1 & -3 & 3 & -9 & 0 \end{array}$$

$$z_1 = 2$$

$$\Rightarrow A(z) = (z - 2)(z^3 - 3z^2 + 3z - 9)$$

$$B(z) = z^3 - 3z^2 + 3z - 9$$

$$B(3) = 27 - 27 + 9 - 9$$

$$\begin{array}{r|rrr|r} & 1 & -3 & 3 & -9 \\ 3 & \downarrow & 3 & 0 & 9 \\ \hline & 1 & 0 & 3 & 0 \end{array}$$

$$z_2 = 3$$

$$\Rightarrow A(z) = (z - 2)(z - 3)(z^2 + 3)$$

$$z^2 + 3 \Rightarrow D = 0 - 12 = -12$$

$$z_{3,4} = \frac{-0 \pm 2\sqrt{3}i}{2} = \pm\sqrt{3}i$$

$$\Rightarrow A(z) = (z - 2)(z - 3)(z - \sqrt{3}i)(z + \sqrt{3}i)$$

References